
Chapter 18 Amino Acid Oxidation and the Production of Urea

Multiple Choice Questions

1. Under which circumstances are amino acids *not* metabolized via oxidative degradation?

- A) Starvation
- B) Plants growing in nutrient-rich soils
- C) Normal protein turnover
- D) A diet rich in proteins
- E) Uncontrolled diabetes

Metabolic fates of amino groups

Page: 695 Difficulty: 2 Ans: B

2. Which of these is *not* a protease that acts in the small intestine?

- A) Chymotrypsin
- B) Elastase
- C) Enteropeptidase
- D) Secretin
- E) Trypsin

Metabolic fates of amino groups

Pages: 697–698 Difficulty: 2 Ans: D

3. In the digestion of protein that occurs in the small intestine, which enzyme is critical in the activation of zymogens?

- A) Enteropeptidase
- B) Hexokinase
- C) Papain
- D) Pepsin
- E) Secretin

Metabolic fates of amino groups

Page: 698 Difficulty: 2 Ans: A

4. Which of the following is a zymogen that can be converted to an endopeptidase that hydrolyzes peptide bonds adjacent to Lys and Arg residues?

- A) Chymotrypsinogen
- B) Pepsin
- C) Pepsinogen
- D) Trypsin
- E) Trypsinogen

Metabolic fates of amino groups

Pages: 698–699 Difficulty: 2 Ans: E

5. In amino acid catabolism, the first reaction for many amino acids is a(n):

- A) decarboxylation requiring thiamine pyrophosphate (TPP).
- B) hydroxylation requiring NADPH and O₂.
- C) oxidative deamination requiring NAD⁺.
- D) reduction requiring pyridoxal phosphate (PLP).
- E) transamination requiring pyridoxal phosphate (PLP).

Metabolic fates of amino groups

Page: 699 Difficulty: 2 Ans: E

6. The coenzyme required for all transaminations is derived from:

- A) niacin.
- B) pyridoxine (vitamin B₆).
- C) riboflavin.
- D) thiamin.
- E) vitamin B₁₂.

Metabolic fates of amino groups

Page: 699 Difficulty: 2 Ans: B

7. The coenzyme involved in a transaminase reaction is:

- A) biotin phosphate.
- B) lipoic acid.
- C) nicotinamide adenine dinucleotide phosphate (NADP⁺).
- D) pyridoxal phosphate (PLP).
- E) thiamine pyrophosphate (TPP).

Metabolic fates of amino groups

Page: 699 Difficulty: 1 Ans: D

8. Transamination from alanine to α -ketoglutarate requires the coenzyme:

- A) biotin.
- B) NADH.
- C) No coenzyme is involved.
- D) pyridoxal phosphate (PLP).
- E) thiamine pyrophosphate (TPP).

Metabolic fates of amino groups

Page: 699 Difficulty: 1 Ans: D

9. Which of the following reactions involving an amino acid *cannot* be catalyzed via a PLP-dependent mechanism?

- A) Hydrolysis
- B) Decarboxylation

- C) Racemization
- D) Transamination
- E) Transimination

Metabolic fates of amino groups**Pages: 700–701 Difficulty: 1 Ans: A**

10. Which of the following is *not* true of the reaction catalyzed by glutamate dehydrogenase?
- A) It is similar to transamination in that it involves the coenzyme pyridoxal phosphate (PLP).
 - B) NH_4^+ is produced.
 - C) The enzyme can use either NAD^+ or NADP^+ as a cofactor.
 - D) The enzyme is glutamate-specific, but the reaction is involved in oxidizing other amino acids.
 - E) α -Ketoglutarate is produced from an amino acid.

Metabolic fates of amino groups**Pages: 700–702 Difficulty: 2 Ans: A**

11. Glutamate is metabolically converted to α -ketoglutarate and NH_4^+ by a process described as:
- A) deamination.
 - B) hydrolysis.
 - C) oxidative deamination.
 - D) reductive deamination.
 - E) transamination.

Metabolic fates of amino groups**Pages: 700–702 Difficulty: 2 Ans: C**

12. The conversion of glutamate to an α -ketoacid and NH_4^+ :
- A) does not require any cofactors.
 - B) is a reductive deamination.
 - C) is accompanied by ATP hydrolysis catalyzed by the same enzyme.
 - D) is catalyzed by glutamate dehydrogenase.
 - E) requires ATP.

Metabolic fates of amino groups**Pages: 700–702 Difficulty: 1 Ans: D**

13. Urea synthesis in mammals takes place primarily in tissues of the:
- A) brain.
 - B) kidney.
 - C) liver.
 - D) skeletal muscle.
 - E) small intestine.

Nitrogen excretion and the urea cycle**Page: 704 Difficulty: 1 Ans: C**

14. Which substance is *not* involved in the production of urea from NH_4^+ via the urea cycle?

- A) Aspartate
- B) ATP
- C) Carbamoyl phosphate
- D) Malate
- E) Ornithine

Nitrogen excretion and the urea cycle

Page: 705 Difficulty: 1 Ans: D

15. Which of these *directly* donates a nitrogen atom for the formation of urea during the urea cycle?

- A) Adenine
- B) Aspartate
- C) Creatine
- D) Glutamate
- E) Ornithine

Nitrogen excretion and the urea cycle

Page: 705 Difficulty: 1 Ans: B

16. Conversion of ornithine to citrulline is a step in the synthesis of:

- A) aspartate.
- B) carnitine.
- C) pyruvate.
- D) tyrosine.
- E) urea.

Nitrogen excretion and the urea cycle

Pages: 705–706 Difficulty: 1 Ans: E

17. In the urea cycle, ornithine transcarbamoylase catalyzes:

- A) cleavage of urea to ammonia.
- B) formation of citrulline from ornithine and another reactant.
- C) formation of ornithine from citrulline and another reactant.
- D) formation of urea from arginine.
- E) transamination of arginine.

Nitrogen excretion and the urea cycle

Pages: 705–706 Difficulty: 1 Ans: B

18. Which of the following statements is *false* in reference to the mammalian synthesis of urea?

- A) Krebs was a major contributor to the elucidation of the pathway involved.
- B) The amino acid arginine is the immediate precursor to urea.
- C) The carbon atom of urea is derived from mitochondrial HCO_3^- .
- D) The precursor to one of the nitrogens of urea is aspartate.
- E) The process of urea production is an energy-yielding series of reactions.

Nitrogen excretion and the urea cycle**Pages: 705–706 Difficulty: 1 Ans: E**

19. Which of the following amino acids are essential for humans?

- A) Alanine
- B) Aspartic acid
- C) Asparagine
- D) Serine
- E) Threonine

Nitrogen excretion and the urea cycle**Pages: 709 Difficulty: 2 Ans: E**

20. If a person's urine contains unusually high concentrations of urea, which one of the following diets has he or she probably been eating recently?

- A) High carbohydrate, very low protein
- B) Very high carbohydrate, no protein, no fat
- C) Very very high fat, high carbohydrate, no protein
- D) Very high fat, very low protein
- E) Very low carbohydrate, very high protein

Nitrogen excretion and the urea cycle**Page: 709 Difficulty: 2 Ans: E**

21. Which of these amino acids can be directly converted into a citric acid cycle intermediate by transamination?

- A) Glutamic acid
- B) Serine
- C) Threonine
- D) Tyrosine
- E) Proline

Pathways of amino acid degradation**Page: 707 Difficulty: 2 Ans: A**

22. Which of these amino acids are *both* ketogenic *and* glucogenic?

- 1. Isoleucine
- 2. Valine
- 3. Histidine
- 4. Arginine
- 5. Tyrosine

- A) 1 and 5
- B) 1, 3, and 5
- C) 2 and 4
- D) 2, 3, and 4

E) 2, 4, and 5

Pathways of amino acid degradation

Page: 711 Difficulty: 2 Ans: A

23. Tetrahydrofolate (THF) and its derivatives shuttle _____ between different substrates.

- A) electrons
- B) H^+
- C) acyl groups
- D) one carbon units
- E) NH_2 groups

Pathways of amino acid degradation

Page: 712 Difficulty: 2 Ans: D

24. Which of the following is *not* a form of the most oxidized state of tetrahydrofolate (THF)?

- F) N^{10} -formyl THF
- G) N^5N^{10} -methenyl THF
- H) N^5N^{10} -methylene THF
- I) N^5 -formyl THF
- J) N^5 -formimino THF

2.

Pathways of amino acid degradation

Page: 713 Difficulty: 2 Ans: C

25. The amino acids serine, alanine, and cysteine can be catabolized to yield:

- A) fumarate.
- B) pyruvate.
- C) succinate.
- D) α -ketoglutarate.
- E) None of the above

Pathways of amino acid degradation

Pages: 715–716 Difficulty: 2 Ans: B

26. Serine or cysteine may enter the citric acid cycle as acetyl-CoA after conversion to:

- A) oxaloacetate.
- B) propionate.
- C) pyruvate.
- D) succinate.
- E) succinyl-CoA.

Pathways of amino acid degradation

Pages: 715–716 Difficulty: 2 Ans: C

27. The human genetic disease phenylketonuria (PKU) can result from:

- A) deficiency of protein in the diet.
- B) inability to catabolize ketone bodies.
- C) inability to convert phenylalanine to tyrosine.
- D) inability to synthesize phenylalanine.
- E) production of enzymes containing no phenylalanine.

Pathways of amino acid degradation**Pages: 717–720 Difficulty: 1 Ans: C**

28. In the human genetic disease maple syrup urine disease, the metabolic defect involves:

- A) a deficiency of the vitamin niacin.
- B) oxidative decarboxylation.
- C) synthesis of branched chain amino acids.
- D) transamination of an amino acid.
- E) uptake of branched chain amino acids into liver.

Pathways of amino acid degradation**Pages: 717, 723 Difficulty: 1 Ans: B**

29. Which of these amino acids are converted to α -ketoglutarate?

- 1. Glycine
 - 2. Glutamate
 - 3. Histidine
 - 4. Arginine
 - 5. Proline
-
- A) 1, 3, and 5
 - B) 2, 3, and 4
 - C) 2, 3, and 5
 - D) 2, 3, 4, and 5
 - E) 3, 4, and 5

Pathways of amino acid degradation**Page: 721 Difficulty: 2 Ans: D**

30. Which of these amino acids are converted to succinyl CoA?

- 1. Isoleucine
 - 2. Valine
 - 3. Methionine
 - 4. Arginine
 - 5. Threonine
-
- A) 2 and 4
 - B) 2, 3, and 4
 - C) 2, 4, and 5
 - D) 1 and 5
 - E) 1, 3, and 5

Pathways of amino acid degradation**Page: 722 Difficulty: 2 Ans: E**

31. Which of these amino acids are converted to oxaloacetate?

1. Asparagine
2. Glutamine
3. Serine
4. Arginine
5. Aspartate

- A) 2 and 4
B) 2, 3, and 4
C) 2, 4, and 5
D) 1 and 5
E) 1, 3, and 5

Pathways of amino acid degradation**Page: 724 Difficulty: 2 Ans: D****Short Answer Questions**

32. Describe (briefly) the role of (a) gastrin, (b) pepsinogen, (c) cholecystokinin, and (d) enteropeptidase in protein digestion.

Ans: (a) The hormone gastrin stimulates secretion of HCl and pepsinogen into the stomach; (b) pepsinogen, after its conversion to pepsin, begins protein degradation in the stomach; (c) the hormone cholecystokinin stimulates the secretion of the pancreatic zymogens chymotrypsinogen, trypsinogen, and procarboxypeptidase into the small intestine; (d) enteropeptidase is a proteolytic enzyme that activates trypsinogen in the small intestine. Trypsin in turn activates chymotrypsinogen, procarboxypeptidases, proelastase, and trypsinogen itself.

Metabolic fates of amino groups**Pages: 697–698 Difficulty: 3**

33. In the treatment of diabetes, insulin is given intravenously. Why can't this hormone, a small protein, be taken orally?

Ans: First, the biological activity of insulin would be destroyed by the low pH of the gastric juice and the proteases that act in the small intestine. Furthermore, even if insulin escaped degradation in the intestine, it would not enter the bloodstream from the intestine; the transport systems in the cells that line the intestinal lumen transport free amino acids, not intact proteins.

Metabolic fates of amino groups**Pages: 697–698 Difficulty: 2**

34. The bacterium responsible for gastric ulcers, *Helicobacter pylori*, survives the acidic pH of the stomach surprisingly well. It does so in part by synthesizing and excreting large amounts of the enzyme urease, which hydrolyzes urea to bicarbonate and ammonia. Explain how these reaction products lead to a less acidic environment for *H. pylori*.

Ans: Bicarbonate is a good buffer. Additionally, any excess bicarbonate that is converted to carbon dioxide generates an additional hydroxide ion that can then neutralize a proton from the stomach. The other reaction product, ammonia, also readily neutralizes a proton, forming NH_4^+ .

Metabolic fates of amino groups

Pages: 697–698 Difficulty: 2

35. Define zymogen and describe the role of one zymogen in protein digestion.

Ans: A zymogen is an inactive form of an enzyme that can be activated by proteolytic cleavage. The pancreatic enzymes pepsinogen, chymotrypsinogen, trypsinogen, and procarboxypeptidases A and B are all inactive forms of proteases, which are activated by proteolytic cleavage after their release into the small intestine.

Metabolic fates of amino groups

Page: 697 Difficulty: 1

36. Transamination reactions are catalyzed by a family of enzymes, all of which require _____ as a coenzyme. In the first step of a transamination, the coenzyme in the aldehyde form condenses with the _____ group of an amino acid to form a(n) _____.

Ans: pyridoxal phosphate (PLP); α -amino; Schiff base (or imine or aldimine)

Metabolic fates of amino groups

Pages: 699–700 Difficulty: 1

37. Draw the structures of reactants and products in the transamination in which glutamate and pyruvate are the starting materials. What cofactor is required for this reaction?

Ans: The correct answer is the reverse of the reaction shown in Fig. 18-4, with $\text{R} = \text{CH}_3$. Pyridoxal phosphate (PLP) is the cofactor.

Metabolic fates of amino groups

Page: 699 Difficulty: 1

38. Give the name and draw the structure of the α -keto acid resulting when the following amino acids undergo transamination with α -ketoglutarate: (a) glutamate; (b) aspartate; (c) alanine.

Ans: (a) α -ketoglutarate; (b) oxaloacetate; (c) pyruvate; see Fig. 18-4.

Metabolic fates of amino groups

Page: 699 Difficulty: 1

39. Describe, by showing the chemical intermediates, the role of pyridoxal phosphate (PLP) in the transamination of an amino acid.

Ans: See the top line of Fig. 18-6.

Metabolic fates of amino groups

Page: 701 Difficulty: 3

40. Describe the roles of glutamine synthetase and glutaminase in the metabolism of amino groups in mammals.

Ans: In tissues that are metabolizing the carbon skeletons of amino acids, the amino groups are transferred by transamination to glutamate, then released as ammonia. Ammonia, which is toxic, is then combined with glutamate to form glutamine; the reaction is catalyzed by glutamine synthetase and requires ATP. Glutamine is moved from the extrahepatic tissues to the liver and kidneys, where the amino group is released from glutamine by glutaminase; the products are glutamate and ammonia. The ammonia delivered in this way to the liver is converted to urea, then excreted.

Metabolic fates of amino groups

Pages: 701–703 Difficulty: 3

41. Show the reaction in which ammonia is formed from glutamate; include any required cofactors and/or intermediates.

Ans: This is the reaction catalyzed by glutamate dehydrogenase; see Fig. 18-7.

Metabolic fates of amino groups

Page: 702 Difficulty: 2

42. Describe the reactions and the role of the glucose-alanine cycle.

Ans: Toxic ammonia formed by amino acid catabolism in muscle is transported to the liver as alanine, which is nontoxic. Alanine is formed in muscle by transamination of pyruvate (formed by glycolysis); glutamate formed by glutamate dehydrogenase is the amino donor. In the liver, alanine is reconverted to pyruvate by transamination. Its amino group is eventually converted to urea, and the pyruvate is converted to glucose by gluconeogenesis in the liver, then exported to muscle.

Metabolic fates of amino groups

Pages: 703–704 Difficulty: 2

43. Why does a mammal go to all of the trouble of making urea from ammonia rather than simply excreting ammonia as many bacteria do?

Ans: When bacteria release ammonia into the surrounding medium, it is diluted to nontoxic levels. The ammonia produced by amino acid catabolism in mammals cannot be sufficiently diluted in the tissues and the blood to avoid accumulating at toxic levels. Urea is much less toxic than ammonia.

Nitrogen excretion and the urea cycle

Page: 704 Difficulty: 2

44. Describe the three different forms of nitrogen by which different organisms dispose of excess nitrogen obtained in the diet. Give examples of organisms that produce these different forms.

Ans: (1) Ammonotelic: release into the surrounding medium as NH_4^+ (bacteria and many marine organisms); (2) Uricotelic: production of uric acid (birds and reptiles); (3) Ureotelic: production and excretion of urea (land-dwelling animals).

Nitrogen excretion and the urea cycle

Page: 704 Difficulty: 2

45. Amino acid catabolism involves the breakdown of 20 amino acids all of which contain nitrogen but have different carbon skeletons. What overall strategy is used to deal with this problem? Illustrate the strategy with two examples.

Ans: Nitrogen is removed by transamination to glutamate. This converts the amino acid to an α -keto acid that either is an intermediate in carbohydrate catabolism or is converted to one. (See Fig. 18-2) Examples are shown in Fig. 18-10.

Nitrogen excretion and the urea cycle

Pages: 697, 706 Difficulty: 2

46. During starvation, more urea production occurs. Explain this observation.

Ans: During starvation, cellular proteins are degraded and their carbon skeletons are oxidized for energy. The first step in amino acid catabolism is removal of the amino groups, which are ultimately excreted as urea.

Nitrogen excretion and the urea cycle

Page: 708 Difficulty: 1

47. Describe (a) the fundamental nutritional problem faced by individuals with genetic defects in enzymes involved in urea formation and (b) two approaches to treatment of these diseases.

Ans: (a) A defect in urea synthesis can result in the formation of toxic blood levels of ammonia from the breakdown of ingested proteins. Thus, it is desirable to limit the intake of amino acids. However, some amino acids are essential for humans (i.e., not biosynthesized) and hence must be ingested in adequate amounts. (b) One approach is to administer compounds that deplete the supply of glycine and glutamine. The replenishment of these amino acids removes ammonia from the blood. Another approach is to administer compounds that allow the liver to bypass the enzyme that is defective in the individual.

Nitrogen excretion and the urea cycle

Pages: 709–710 Difficulty: 2

48. Explain why one might provide high concentrations of the amino acid Arg to patients who have a deficiency in one of the first three enzymes of the urea cycle.

Ans: The Arg is converted efficiently to urea and ornithine since the last enzyme of the urea cycle is still functioning normally. Thus, one produces higher than normal concentrations of ornithine, which is required for the rest of the cycle to work. That is, you increase the flux through the pathway by providing higher concentrations of the intermediates.

Nitrogen excretion and the urea cycle

Pages: 709–710 Difficulty: 2

49. Name four amino acids that can be converted directly (in one step) into pyruvate or a citric acid cycle intermediate, and name the intermediate formed from each.

Ans: (1) aspartate; oxaloacetate; (2) glutamate; α -ketoglutarate; (3) alanine; pyruvate; (4) serine;

pyruvate. (Order is not important.)

Pathways of amino acid degradation**Pages: 710–711 Difficulty: 1**

50. Name one amino acid whose oxidation proceeds via the intermediate shown: (a) pyruvate; (b) oxaloacetate; (c) α -ketoglutarate; (d) succinyl-CoA; (e) fumarate.

Ans: Possible answers are: (a) alanine, tryptophan, glycine, serine, cysteine; (b) aspartate, asparagine; (c) glutamate, glutamine, arginine, histidine, proline; (d) isoleucine, threonine, methionine, valine; (e) phenylalanine, tyrosine.

Pathways of amino acid degradation**Pages: 710–711 Difficulty: 2**

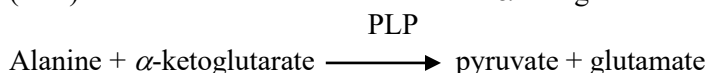
51. Degradation of amino acids yields compounds that are common intermediates in the major metabolic pathways. Explain the distinction between glucogenic and ketogenic amino acids in terms of their metabolic fates.

Ans: The glucogenic amino acids are those that are catabolized to intermediates that can serve as substrates for gluconeogenesis: pyruvate and any of the four- or five-carbon intermediates of the citric acid cycle. Ketogenic amino acids are catabolized to yield acetyl-CoA or acetoacetyl-CoA, the precursors for ketone body formation.

Pathways of amino acid degradation**Pages: 710–711 Difficulty: 2**

52. There are bacteria for which alanine can serve as the chief energy source; they oxidize the carbon skeleton of this amino acid, thereby generating ATP. Describe the first step in alanine degradation; show any cofactors that participate.

Ans: The first step is removal of the amino group by transamination, in which pyridoxal phosphate (PLP) serves as an essential cofactor and α -ketoglutarate is the usual amino group acceptor:

**Pathways of amino acid degradation****Pages: 715 Difficulty: 2**

53. If you received a laboratory report showing the presence of a high concentration of phenylalanine and its metabolites in the urine of a patient, what disease would you suspect? What defect(s) in metabolism account(s) for the accumulation of phenylalanine in such patients?

Ans: Phenylketonuria, the disease that results from a defect in either phenylalanine hydroxylase or the enzyme that catalyzes regeneration of tetrahydrobiopterin.

Pathways of amino acid degradation**Pages: 718–720 Difficulty: 2**

54. Suppose you are responsible for formulating the diet for a 4-year-old boy with phenylketonuria. How do you decide what kind and amount of protein to include in the diet?

Ans: A growing child needs a certain amount of phenylalanine and tyrosine for the synthesis of new proteins. Because an excess of phenylalanine and tyrosine is toxic in phenylketonurics, the diet should include just enough of these amino acids in protein to supply these needs, but not enough so that phenylketones begin to accumulate in the blood and urine. If the defect causing phenylketonuria lies in the enzyme that regenerates tetrahydrobiopterin, l-3,4-dihydroxyphenylalanine and 5-hydroxytryptophan must also be supplied in the diet. Adding tetrahydrobiopterin to the diet will not help because the compound is unstable and cannot cross the blood-brain barrier.

Pathways of amino acid degradation

Pages: 718–720

Difficulty: 2

55. Diagram the degradative pathway from proline to an intermediate of either glycolysis or the citric acid cycle. Show structures of intermediates and indicate where cofactors are involved.

Ans: The shortest pathway is from proline to Δ^1 -pyrroline-5-carboxylate, glutamate semialdehyde, glutamate, and α -ketoglutarate. See Fig. 18-26.

Pathways of amino acid degradation

Pages: 721–722

Difficulty: 3